Skills Summary

Sign and Conjugate Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Products and the Sign of i Squared

Rule: $i^2 = -1$, so every i^2 term flips sign when it becomes real.

$(a+bi)^2 = a^2 - b^2 + 2abi \qquad (a+bi)(a-bi) = a^2 + b^2$

Notice the two signs pull opposite ways: a square *subtracts* b^2, a conjugate product *adds* it. Both come from the same $i^2 = -1$.

Example: Expand $(4-3i)^2$

Step 1. A square is a product of two identical binomials, so all four products must be written — the middle term is the one that gets dropped.

Step 2. $16 - 12i - 12i + 9i^2$, and $9i^2 = -9$, so the real part is $16 - 9$ and the imaginary part is -24.

$\Rightarrow$ $\boldsymbol{7 - 24i}$

Try it: Expand $(2+5i)^2$

check: $\boldsymbol{-21 + 20i}$

Skill 2 — Distributing a Subtraction Sign

Rule: a minus sign in front of a complex number reverses *both* of its parts.

$(a+bi) - (c+di) = (a-c) + (b-d)i$

Rewrite the subtraction as an addition first — $-(c+di)$ becomes $-c-di$ — and the sign error has nowhere left to hide.

Example: Simplify $(7+3i) - (2-9i)$

Step 1. Nothing needs multiplying; the only risk is the sign, so rewrite the second number with both parts reversed before collecting.

Step 2. $7 + 3i - 2 + 9i$, so the real parts give $7-2$ and the imaginary parts give $3+9$.

$\Rightarrow$ $\boldsymbol{5 + 12i}$

Try it: Simplify $(4-6i) - (-3+8i)$

check: $\boldsymbol{7 - 14i}$

Skill 3 — Choosing the Right Conjugate

Rule: to clear an i from a denominator, multiply above *and* below by the conjugate of the denominator — the same number with the sign of its imaginary part reversed.

$(c+di)(c-di) = c^2 + d^2$, always real and always a *sum*.

Two ways this goes wrong: using the denominator itself instead of its conjugate, and multiplying only the bottom by the conjugate.

Example: Simplify $\dfrac{25}{4+3i}$

Step 1. An i downstairs means the quotient is not in standard form, so multiply top and bottom by the conjugate $4-3i$.

Step 2. Bottom: $16+9=25$. Top: $25(4-3i)$. The two 25s cancel, leaving the real part 4 and the imaginary part -3.

$\Rightarrow$ $\boldsymbol{4-3i}$

Try it: Simplify $\dfrac{34+17i}{4-i}$

check: $\boldsymbol{7+6i}$

(!) Watch out: when you check someone else's work, test the *denominator* first — $9-4$ where $9+4$ belongs is the single most common conjugate slip, and every later line of their solution will look tidy.